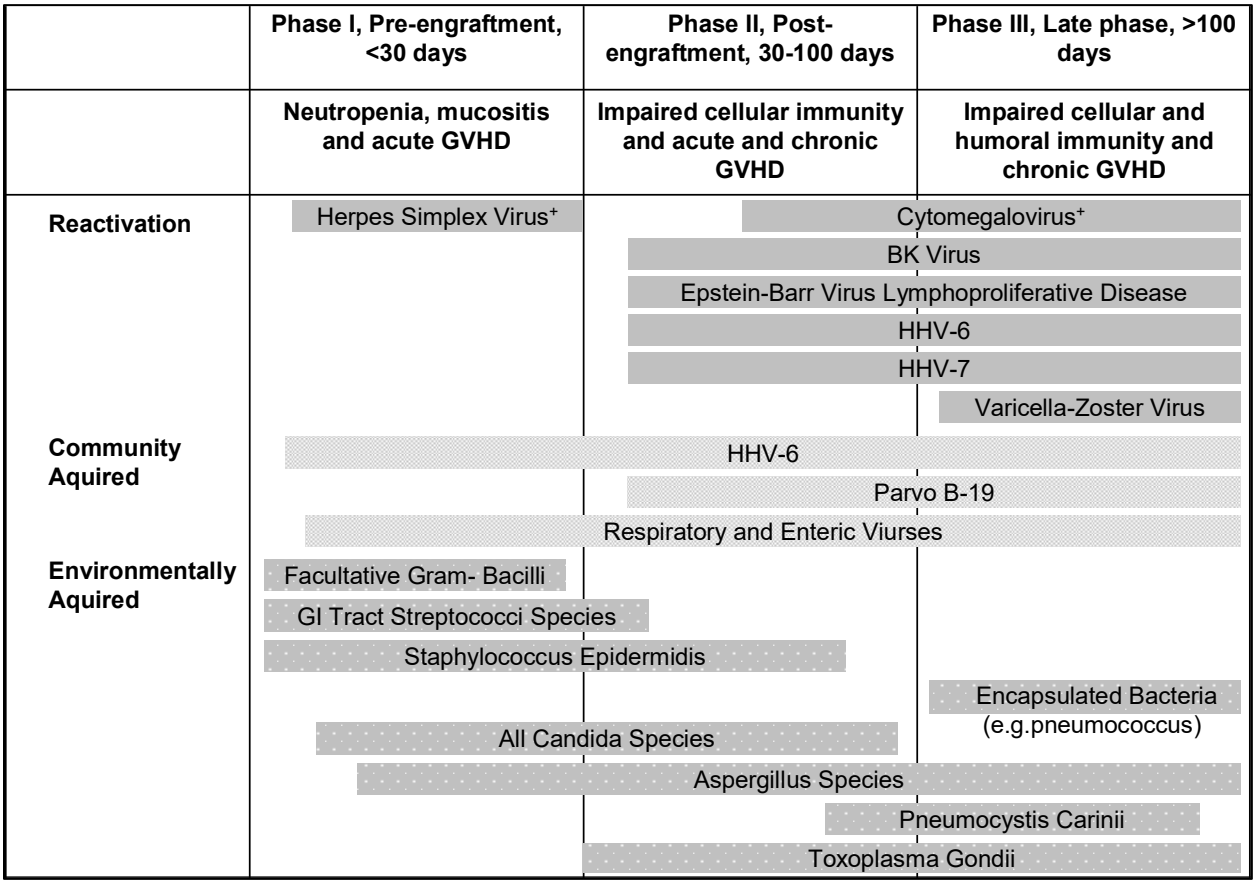
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INFECTION PROPHYLAXIS AND TREATMENT FOR ADULT AND PEDIATRIC BONE MARROW TRANSPLANT RECIPIENTS

I. **Purpose:** BMT patients are at risk for infections in the post-transplant period.

II. **Scope:** This guideline outlines the routine infection prophylaxis for at risk patients.

III. **Guideline:**



+ Primarily among persons who are seropositive before transplant.

Table 1. Prevention of PJP and Toxoplasma

Pathogens: *Pneumocystis jiroveci* and *Toxoplasma**

Indication	First choice	Alternatives
All allogeneic and autologous transplant recipients with additional immunosuppression (e.g., corticosteroids, T-cell depletion, ongoing cancer therapy) Time: Start once ANC >1,000 and PLT >50,000 but no earlier than day 30. Duration (Allo): Stop at 6mo or until off IS^ Duration (Auto): 3-6 months	Adult, PJP only: **TMP-SMX** 1 DS tab PO BID, 2 days per week (i.e., M & Th) Adult, PJP and *Toxoplasma* IgG positive: **TMP-SMX** 1 DS tab PO BID, 3 x/week Peds: **TMP-SMX** 2.5 mg/kg TMP PO BID, 2 days per week (max: 160 mg TMP/dose)	Preferred alternative for PJP prophylaxis AND IgG Positive for *Toxoplasma*: *Adult:* **Atovaquone** (Mepron) 1500 mg PO daily – take with food *Pediatric:* **Atovaquone** – take with food 1-3 mo or >24 mo: 30 mg/kg/day PO Daily; 4-24 mo: 45 mg/kg/day (max: 1500 mg/dose)
*All allogeneic HCT recipients should have a screening *Toxoplasma* IgG prior to transplantation. Patients who are *Toxoplasma* IgG positive and unable to receive TMP/SMX prophylaxis should have weekly *Toxoplasma* PCR performed for a minimum of 8 weeks post-transplant and until CD4 count >200 cells/mm3 (whichever comes later) and during periods of intensive immunosuppression for GVHD. In patients with repeated negative PCRs and prolonged immunosuppression, may consider reducing frequency of or stopping routine monitoring. ^Patients with GvHD should be restarted or continue to receive PJP/ *Toxoplasma* prophylaxis TMP-SMX : Trimethoprim-sulfamethoxazole (Bactrim) DS tab: double strength tablet (Trimethoprim 160 mg-sulfamethoxazole 800 mg)		Alternative for PJP prophylaxis AND IgG Negative for *Toxoplasma*: *Adult:* **Pentamidine** 300 mg inh once monthly OR **Pentamidine** 4 mg/kg IV once monthly (max dose 300mg; if unable to tolerate inhaled; dosed using actual body weight) *Pediatric:* *<5 years:* **Pentamidine** 9 mg/kg inh once monthly (max: 300 mg) OR **Pentamidine** 4 mg/kg IV once monthly (max dose 300mg; if unable to tolerate inhaled; dosed using actual body weight) *≥5 years:* **Pentamidine** 300 mg inh once monthly OR **Pentamidine** 4 mg/kg IV once monthly max dose 300mg;if unable to tolerate inhaled; dosed using actual body weight)
		Alternative for PJP prophylaxis AND IgG Negative for *Toxoplasma*: *Adult:* **Dapsone** 100 mg PO daily (G6PD screening recommended for all patients; may be used in patients with mild/moderate hypersensitivity to sulfa, consult allergy for more severe reactions) *Pediatric:*

	Dapsone 2 mg/kg/dose PO daily (max: 100 mg/dose) (G6PD screening recommended for all patients; caution in patients with hypersensitivity to sulfa; dose-dependent risk of methemoglobinemia independent of G6PD status)

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Table 2. Prevention of Fungal (Yeast and Mold) Infections			
Indication	**First Choice**	**Alternative**	**Duration**
Low-risk (excludes cord, haploidentical, receipt of post-transplant cyclophosphamide, pharmacologic or ex-vivo T-cell depletion, mismatched) **Starting on admission**	Adults: **Fluconazole** 200 mg PO/IV Daily Pediatric: **Fluconazole** 3 mg/kg PO/IV Daily (max: 200 mg)	If AST/ALT/T. Bili >3x ULN or otherwise deemed clinically significant, consider: *Adults:* **Micafungin** 50 mg IV q24h *Pediatric:* **Micafungin** 3-4 mg/kg IV q24h (max: 50 mg)	Auto: Until ANC >1000 x3 days Allo: Until day +100, off immunosuppression or on broader antifungal
Scenario 1: All cord, haploidentical, recipients of post-transplant cyclophosphamide, mismatcherecipients, or ATG, thymoglobulin, and alemtuzumab conditioning regimens Scenario 2: Patients with prolonged neutropenia (≥30 days) entering transplant **Starting on admission**	Adults: **Micafungin** 100 mg IV q24h, then **Posaconazole** delayed-release tabs 300 mg PO/IV◆ daily Pediatric: **Micafungin** 5 mg/kg IV q24h, then **Posaconazole** delayed-release tabs: <10 kg: 50mg PO BID 10 - <20 kg: 100mg PO BID 20 - <35 kg: 200mg PO BID ≥35 kg: 300mg PO BID αTransition to posaconazole can be delayed if AST/ALT/T. Bili >3x ULN or otherwise deemed clinically significant	If insurance does not cover posaconazole: *Adults:* **Voriconazole** 200 mg PO/IV◆ BID starting day +5-10α *Pediatric:* **Voriconazole** 8 mg/kg PO/IV◆ BID (TID if ≤8 years old) starting day +5-10 α Isavuconazole is not recommended at this time due to reports of breakthrough infection. May be considered if >15 years AND >50 kg AND either QTc>500, hepatotoxicity to vori/posa, or insurance does not cover vori/posa: **Isavuconazole** 372 mg PO/IV◆ daily	Scenario 1: Continue until day +100 Note: If still on immunosuppression when mold-active agent discontinued, may switch to fluconazole (same dosing as above) until off immunosuppression. Scenario 2: Until ANC >1000 x3 days
Allogeneic patients with GvHD or engraftment syndrome receiving additional immunosuppression (systemic steroids ≥0.25 mg/kg, infliximab, ruxolitinib, etc.)	Adults: **Posaconazole** delayed-release tabs 300 mg PO/IV◆ daily Pediatric: **Posaconazole** delayed-release tabs: <10 kg: 50mg PO BID 10 - <20 kg: 100mg PO BID 20 - <35 kg: 200mg PO BID ≥35 kg: 300mg PO BID	If concern for liver GvHD (AST/ALT/T. Bili >3x ULN): *Adults:* **Micafungin** 100 mg IV q24h *Pediatric:* **Micafungin** 5 mg/kg IV q24h (max: 100 mg) If insurance does not cover posaconazole: *Adults:* **Voriconazole** 200 mg PO/IV◆ BID *Pediatric:* **Voriconazole** 8 mg/kg PO/IV◆ BID (TID if ≤8 years old) Isavuconazole is not recommended at this time due to reports of breakthrough infection. May be considered if >15 y/o AND >50 kg AND either QTc>500, hepatotoxicity to vori/posa, or insurance does not cover vori/posa. **Isavuconazole** 372 mg PO/IV◆ daily	Discontinue when off additional GvHD-related immunosuppression (or <10 mg/day prednisone (adults) or 0.15 mg/kg/day (pediatric) for >30 days). See footnote^ for prophylaxis duration after other GvHD-related immunosuppression.
***Trough levels of voriconazole, isavuconazole, and posaconazole should be drawn after 5-7 days. Refer to guideline on therapeutic drug monitoring of antifungal agents ^ Alemtuzumab and Anti-thymocyte globulin (6 months); Infliximab, Tocilizumab, Etanercept, Adalimumab, Basiliximab, Vedolizumab (3 months) ECP, Ruxolitinib (1 month)		◆IV azole preferred over micafungin in high risk patients who are NPO with an expected duration of treatment >5 days. Change to PO azole when appropriate.	

Table 3. Prevention of Bacterial Infections

Pathogen	Indications	First choice	Duration
Bacterial prophylaxis	All Autologous and Allogeneic transplant recipients <u>starting D+1</u>	<u>Adult:</u> **Levofloxacin** 500 mg PO/IV daily <u>Pediatric:</u> *<5 years:* **Levofloxacin** 10 mg/kg PO/IV BID (max: 375 mg/dose) *≥5 years:* **Levofloxacin** 10 mg/kg PO/IV daily (max: 500 mg/dose) <u>Alternative for FQ Allergy/Intolerance:</u> *Adult:* **Cefpodoxime** 200 mg PO BID *Pediatric:* **Cefpodoxime** 5 mg/kg PO BID (max: 200 mg/dose)	Pre-engraftment prophylaxis, continue until: 1. Fever and Neutropenia (i.e., broader antimicrobial such as cefepime, etc.) 2. Engraftment◆
	Patients to be started on levofloxacin as prophylaxis post-transplant for the following indications: 1. Acute GvHD with <u>GI involvement</u> 2. Pancytopenia 2/2 graft failure		GI GvHD: Continue until normal PO intake Pancytopenia: Until count recovery
Encapsulated organisms	Recipients with chronic GvHD, s/p splenectomy, functionally asplenic, or sickle cell disease	<u>Adults:</u> **Penicillin VK** 500 mg PO BID <u>Pediatric:</u> *≥10 years:* **Penicillin VK** 500 mg PO BID *3-10 years:* **Penicillin VK** 250 mg PO BID *<3 years* **Penicillin VK** 125 mg PO BID <u>Alternative (Penicillin allergy):</u> *Adults:* **Azithromycin** 250 mg PO daily OR **TMP-SMX** 1 DS PO daily *Peds:* **Azithromycin** 5 mg/kg PO daily (max: 250 mg) OR **TMP-SMX** 2.5 mg TMP/kg BID, 2 days per week (max: 160 mg TMP/dose)	1. Off Immunosuppression for 1 month and asymptomatic 2. <u>Hold while on levofloxacin</u> <u>Lifelong if:</u> 1. s/p splenectomy 2. extensive cGvHD
Immunoglobulin Supplementation	IgG <400 mg/dL	**IVIG** 0.4 g/kg/dose based on ideal body wt	Monitor monthly until IgG >400 for 2 months

◆Absolute neutrophil count ≥1000/mm^3 for 3 consecutive days

Table 4. Prevention of Viral Infections			
Pathogen	**Indication**	**First Choice**	**Duration**
Herpes Simplex Virus (HSV)	All seropositive recipients (starting day 0)	Adult: **Acyclovir*** 400 mg PO BID or 200 mg IV q12h Pediatric: *≥6 years:* **Acyclovir*** 400 mg PO BID or 200 mg IV q12h or 2.5 mg/kg IV q12h *<6 years:* **Acyclovir*** 200 mg PO BID or 2.5 mg/kg IV q12h Alternative (Acyclovir allergy) - Famciclovir *Adults:* **Famciclovir** 250 mg PO BID	1. Stop day +30 if only HSV 2. Stop at 1-year if VZV+ and HSV+ (recipient) 3. Continue indefinitely while receiving treatment for GvHD
Cytomegalovirus (CMV)	**All seropositive** allogeneic recipients (starting day +10) until day +100 •	Adult: **Letermovir^** 480 mg PO/IV daily (240 mg daily with cyclosporine) Pediatric: *≥30 kg:* **Letermovir^** 480 mg PO/IV daily (240 mg daily with cyclosporine) *15 kg to <30 kg:* **Letermovir^** 240 mg PO daily (120 mg daily with cyclosporine) 120 mg IV daily (60 mg daily with cyclosporine) *7.5 kg to <15 kg:* **Letermovir^** 120 mg PO daily (60 mg daily with cyclosporine) 60 mg IV daily (30 mg daily with cyclosporine) <7.5 kg: *Consult Pediatric Infectious Disease Pharmacist* *CONTINUE ACYCLOVIR WHILE ON LETERMOVIR FOR HSV COVERAGE.* ***Letermovir tablets may be split and/or crushed*** Outpatient insurance coverage for letermovir should be verified prior to initiation. *All other uses of letermovir require approval by transplant infectious diseases.*	1. Continue until day +100; may extend letermovir beyond 100 days in patients with GvHD requiring significant ongoing immunosuppression or with suspicion for delayed recovery of T-cell mediated immunity 2. Should be stopped for patients receiving **ganciclovir**, **valganciclovir**, or **foscarnet**. Continue in patients receiving cidofovir regardless of cidofovir dose. 4. 3. No recipient seronegative patients should be started on letermovir regardless of donor status
Varicella-Zoster Virus (VZV)	All recipients with a history of chickenpox, zoster, or positive serology, starting day +30 Pediatric patients with a history of VZV vaccine who are seropositive pre-transplant DO NOT need therapy.	See above HSV section	1. Until +1 year and off immunosuppression 2. Continue indefinitely while receiving treatment for GvHD

*(Val)Acyclovir/Famciclovir should be held if patient is on ganciclovir, valganciclovir, foscarnet or full dose cidofovir but should continue if patient is on Letermovir
^Letermovir should be held if patient is being treated for CMV (see flowchart); Letermovir should be stopped after 2 separate CMV reactivations (PCR >3000) requiring treatment.

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EBV Reactivation Screening Guidelines for Stem Cell Transplant Recipients

- Guidelines to be followed for high risk patients only

Screening Guidelines

- Send quantitative EBV PCR every other week (plasma assay)
- Start testing on day +7 (adults) or with conditioning for high risk pediatric pts (i.e., recent alemtuzumab) and continue through D+100

Reactivation Guidelines

- *Consider* starting treatment with rituximab if EBV copy number >1000 IU/mL on 2 consecutive tests or if symptomatic (fever, lymphadenopathy)
- If copy number >1000 IU/mL
  - Consider CT of chest/abdomen/pelvis
  - Request EBV serostatus on the donor
  - Start **Rituximab** (dose for all patients = 375 mg/m^2); *additional doses should be given if EBV PCR remains positive 1 week after Rituximab is given (i.e., continue rituximab until PCR negative and asymptomatic)*
  - Recommend reduction in immunosuppression if clinical situation allows
  - Ganciclovir treatment is not recommended
- If EBV reactivation leads to PTLD
  - Continue **Rituximab** weekly with consideration for additional chemotherapeutic agents (CHOP based therapy)
  - If donor is seropositive, consideration should be given to donor lymphocyte infusion
  - CNS disease, consideration can be given to intrathecal **Rituximab** (Dose = 12-50 mg)
  - Recommend reduction in immunosuppression if clinical situation allows

Treatment of Toxoplasmosis

If a patient develops new CNS, new neurologic symptoms, or severe sepsis/septic shock concerning for infection, recommend sending *Toxoplasma* PCR at that time. If *Toxoplasma* IgG positive and unable to receive sulfamethoxazole-trimethoprim, perform weekly PCR for *Toxoplasma*. If PCR positive or strong clinical suspicion, treat as listed below.

Indication	First Choice	Alternative for sulfa allergy
* Positive *Toxoplasma* quantitative plasma PCR	Adult: **Sulfamethoxazole-trimethoprim 5mg/kg IV/PO (trimethoprim component) q8 – 12h** **OR** **Pyrimethamine** 200 mg PO loading dose then 75 mg PO daily + **Sulfadiazine** 1 g PO QID + **Leucovorin** 25 mg daily† Pediatric: **Sulfamethoxazole-trimethoprim 5mg/kg IV/PO (trimethoprim component) q8 – 12h** **OR** **Pyrimethamine** 1 mg/kg/dose PO BID (max: 50 mg/dose) x2 days, then 1 mg/kg daily (max: 75 mg/dose) + **Sulfadiazine** 50 mg/kg/dose PO QID (max: 1 g/dose) + **leucovorin** 25 mg daily†	**Pyrimethamine** + **Leucovorin** as per first choice + one of the following: **Clindamycin** IV/PO 1,800 – 2,700 mg/day divided q6-8h OR **Dapsone** 100 mg PO daily OR **Azithromycin** 1,250 mg PO daily†

† Treat for minimum of 7 days after PCR has become negative

* Consult infectious diseases

IV. **Diagnostic Testing Guidelines**

a. In general, quantitative viral PCR testing (e.g., CMV) does not provides clinically usefully information when repeated more frequently than weekly. In patients with low positive HHV-6 plasma PCR assays (<2500 copies/mL), more frequent repeat testing is indicated as a rising titer may result in earlier institution of treatment. In patients for whom viral monitoring is required, recommend serial testing on Mondays.
b. While case reports have demonstrated rare disease associated with HHV-7 (e.g., encephalitis), the significance of HHV-7 viremia (5%-57% of patients at least one-time point) is unknown. Measurement of HHV-7 for undifferentiated fever is unlikely to be clinically useful.
c. HHV-8 is primarily associated with Kaposi's sarcoma (0.5% of post HSCT patients) and should not be measured routinely.
d. Limited studies on the use of quantitative PCR for CMV on stool have been conducted, the assay is not standardized, and clinical utility is unknown.
e. Weekly screening galactomannan levels in patients on prophylaxis with voriconazole, posaconazole, isavuconazole, or micafungin will <u>not</u> be sent. Symptom based testing would be appropriate.

V. HHV-6 Guidelines

a. HHV-6 infection may result in encephalitis after allogeneic stem cell transplant and has been associated with delayed engraftment, pneumonitis, fever and rash. Asymptomatic reactivation occurs in about 50% of patients. The following principles guide our approach to HHV-6.
    i. HHV-6 viremia should not be routinely monitored
    ii. In patients in which there is clinical suspicion for HHV-6 related disease (e.g., encephalitis early after transplant, delayed engraftment, rash and fever without other explanation) a quantitative plasma PCR for HHV-6 should be obtained.
    iii. Treatment with foscarnet, ganciclovir (post-engraftment with adequate blood counts) or cidofovir should be considered if:
        1. Single value >5,000 copies/mL
        2. Doubling of HHV-6 PCR on serial measurement and >2500 copies/mL (serial measurement should be performed as soon as positive titer received)
        3. Transplant ID consultation recommended if treatment considered
        4. Patients at high risk for graft failure or delayed engraftment (cord blood transplants, low dose grafts, HLA mismatched grafts) may benefit most from treatment.

VI. Adenovirus Guidelines

a. Adenovirus infection may result in severe respiratory disease, hepatitis, and colitis after allogeneic stem cell transplant and disease has been associated with high rates of mortality. The incidence of adenovirus infections is higher in pediatric patients (20-26%) undergoing HSCT than in adults (9%). <u>The following principles guide our approach to Adenovirus:</u>
b. Adenovirus serum PCR should be monitored weekly in high risk patients starting on day +7 (adults) or with conditioning for high risk pediatric pts (i.e., intermediate or distal alemtuzumab)
    i. Routine monitoring is not recommended in all other patients
c. In patients in which there is a clinical suspicion for adenovirus related disease (e.g., severe diarrhea or pneumonia, delayed engraftment, etc.) or RPAN or GI PCR positive for adenovirus, a quantitative plasma PCR for adenovirus should be obtained.
    i. Note: RPAN and GI PCR test for different subtypes of adenovirus and may produce discordant results
d. Treatment should be considered if:
    i. Single positive PCR value or doubling of adenovirus PCR on serial measurements
    ii. Diarrhea with GI PCR positive for adenovirus in the absence of alternative etiology

iii. Pneumonia with RPAN PCR positive for adenovirus in the absence of alternative etiology

e. *Consultation to transplant ID* is highly recommended to assist in treatment decision

f. Patients at high risk for graft failure or delayed engraftment (cord blood transplants, low dose grafts, HLA mismatched grafts) may benefit most from treatment.

g. Treatment:

i. Adult: **Cidofovir** 5 mg/kg IV weekly x2 doses then every other week with prehydration and oral **probenecid** 2 g, 2 hours prior to cidofovir, and 1 g 1 & 4 hours after completion of cidofovir.

ii. Pediatric: **Cidofovir** 5 mg/kg IV weekly x2 doses then every other week with prehydration and oral **probenecid** 30 mg/kg (min: 250 mg; max: 2 g) 2 hours prior to cidofovir dose and 15 mg/kg (min: 250 mg, max: 1 g) 1 & 4 hours after completion of cidofovir. OK to crush tablet and mix with food or water prior to administration if needed.

VII. Strongyloides Guidelines

a. Strongyloides may result in severe disseminated disease due to organism hyperinfection resulting in septic shock, Gram negative meningitis, and death following allogeneic stem cell transplant. Exposure is common in tropical regions of the world and southern U.S. and infection may occur in patients with otherwise unremarkable, distant, travel history.

b. Strongyloides IgG should be checked prior to transplant

c. All seropositive recipients should receive ivermectin 200mcg/kg PO daily on days 1, 2, 8, and 9 (first dose of ivermectin being day 1) prior to transplant

VIII. References

a. Guidelines for Preventing Infectious Complications among Hematopoietic Cell Transplant Recipients: A Global Perspective. Biology of blood and marrow transplantation: journal of the American Society for Blood and Marrow Transplantation. 2009;15(10):1143-1238.

b. Levine JE and Ferrara JLM. Practical considerations of immune reconstitution following bone marrow transplantation. In Current Therapy in Allergy, Immunology, and Rheumatology, 6th edition, eds. Lichtenstein LM, Busse WW, Geha RS. Mosby Press, St. Louis, 2003.

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e. Marty FM, Ostrosky-Zeichner L, Cornely OA, et al; for the VITAL and FungiScope Mucormycosis Investigators. Isavuconazole treatment for mucormycosis: a single-arm open-label trial and case-control analysis. Lancet Infect Dis. 2016;16:828-837

Authors:

Gianni Scappaticci, PharmD, David Frame, PharmD, Sam Aitken, PharmD, Gregory Eschenauer, PharmD

Reviewers:

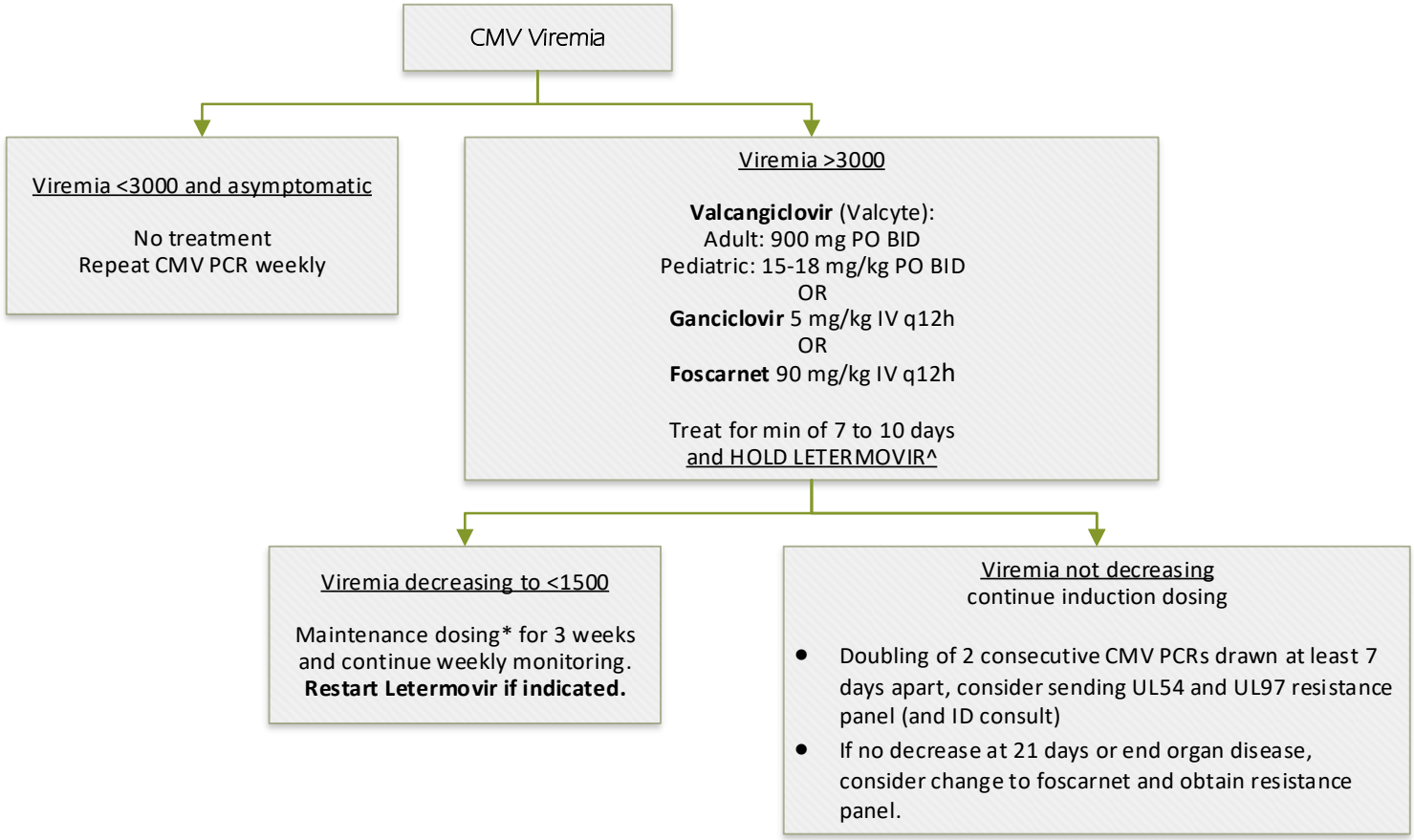
Bone Marrow Transplant

Transplant Infectious Diseases

Additional approvers:

CPC 2/2019

Figure 1: CMV Viremia detected in patient undergoing preemptive monitoring



* Maintenance dosing- **Valganciclovir** 900 mg PO daily, **Ganciclovir** 5 mg/kg IV q24h, **Foscarnet** 90 mg/kg IV daily
Acyclovir should be held if patient is on ganciclovir, valganciclovir, foscarnet, or full dose cidofovir but should continue if patient is on Letermovir

^ Letermovir should be stopped after 2 separate CMV reactivations (PCR >3000) requiring treatment. In patients on letermovir who do not develop GvHD, letermovir should be stopped at day +100.

Table 6. Goal trough levels for voriconazole, isavuconazole, and posaconazole (ordered 5-7 days after starting)

Antifungal	Prophylaxis	Treatment
Voriconazole	>1 mcg/mL	>1 mcg/mL
Isavuconazole	>1 mcg/mL	>1 mcg/mL
Posaconazole	>700 ng/mL	>1,250 ng/mL

Last revised: 03/2025	**Antimicrobial Stewardship Subcommittee:** 02/2025	**P&T:** 03/2025

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	Citation
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3	Marty FM, Ljungman P, Chemaly RF, et al. Letermovir prophylaxis for cytomegalovirus in hematopoietic-cell transplantation. N Engl J Med. 2017;377(25):2433-2444.

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Revision History

Date	Revision
02/2021	Adjusted letermovir criteria
01/2022	Adjusted pediatric acyclovir dosing
3/2025	Added Strongyloides screening, updated letermovir criteria, updated antifungal prophylaxis recommendations